

QUESTIONS, LABELED BY WHETHER REASONING WAS NEEDED

■ CoT-necessary (right with CoT, wrong without)

■ CoT-unnecessary (right both ways)

OPENTHINKER3-7B, VS ITS NON-REASONING PARENT, LAYER 19, CONTEXT → ANSWER MAP

**Predictability of the answer state,
by whether the question needed reasoning**

Held-out R^2 vs corpus mean ↑

